Project 4 | Exercises 1 and 2

Control-based RRT and KPIECE1 - checkpoint

We solve the assigned pendulum swing-up and car navigation problems with OMPL control planners.
The pendulum uses SO(2) x [-10,10] and torque limits 3, 5, and 10. The car uses SE(2) x [-2,2],
including reverse motion; angular velocity and acceleration are each bounded by +/-1.5.
Both planners use identical problem definitions for each system. All figures use actual seed-1 solutions.

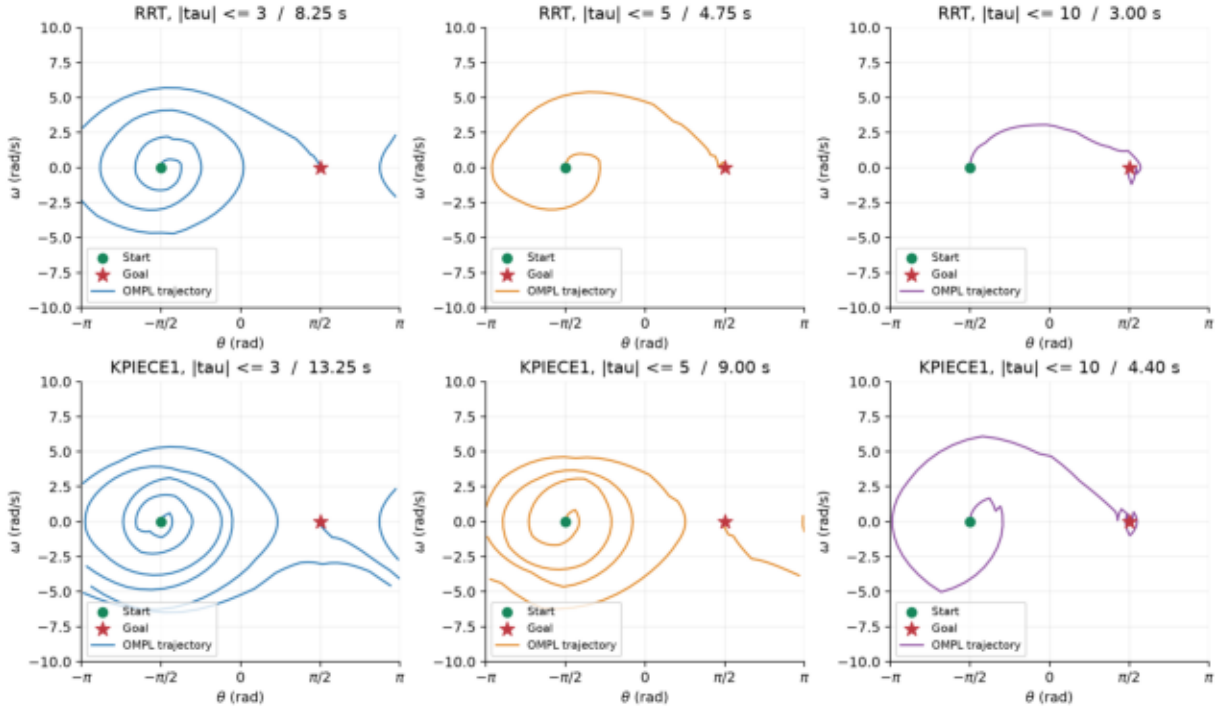
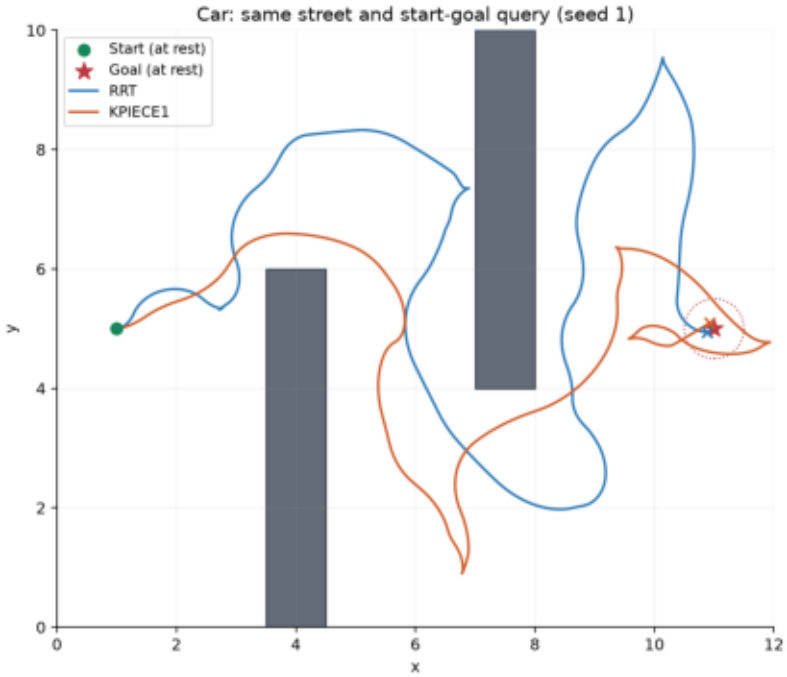


Figure 1. Phase-space trajectories. Times are simulated motion durations; angle seams are broken.



What the paths show

At torque 3, repeated swings build energy. For these examples, RRT motion durations are 8.25, 4.75, and 3.00 s as the torque bound increases. KPIECE1 durations are 13.25, 9.00, and 4.40 s. These are illustrative paths, not optimized motions or a statistical planner comparison.

Figure 2. The 12 x 10 street has two alternating rectangular barriers: (3.5,0,1,6) and (7,4,1,6). A point car starts at (1,5,0,0) and targets (11,5,0,0). Both paths go around the barriers. Crosses mark final positions; the dotted circle is a workspace envelope of the full goal region.

Goal radii in the compound state metrics are 0.1 for the pendulum and 0.5 for the car. Exact solutions reach these neighborhoods; they do not imply exact arrival at zero velocity.

Verification

All 8 configurations succeeded for seeds 1, 2, and 3 (24/24). All three CTest model/collision tests passed.
Independent SciPy replay at <=2.5 ms checked dynamics, bounds, obstacles, and goal arrival; max local error 1.2e-09.
Integration step: 5 ms; planning propagation step: 50 ms. RG-RRT and benchmarking remain unimplemented.